Ethical Approval Application Form

Version 1

Effective date: 1st January 2026

1. ADMINISTRATION DETAILS

Researcher: S M K I Wijethunga

School/Research Centre/Programme (as applicable): Kandy Innovation Centre (KIC), National Institute of Business Management (NIBM) BSc (Hons) Computer Science & Artificial Intelligence

Title of Project: Forecasting Sri Lankan Tourist Arrivals via Hybrid LSTM–XGBoost Ensemble Architectures Under Exogenous Shock Conditions

Supervisor/Research Centre Director/Head of Department: _____

Date: 12/07/2026

Type of research			
Undergraduate	Postgraduate	Staff member	
✓			

There is an obligation on the lead researcher to bring to the attention of the School Ethics Committee any issues with ethical implications not clearly covered by this application form.

2. APPLICATION FORM CHECKLIST

Please complete the ethics application form below and provide additional information as attachments.

My application includes the following documentation:	**INCLUDED (mark as YES)**	**NOT APPLICABLE (mark as N/A)**
Recruitment advertisement		N/A
Participant Information Leaflet		N/A
Participant Informed Consent form		N/A
Questionnaire/Survey		N/A
Interview/Focus Group Questions		N/A
Debriefing material		N/A
Evidence of approval to gain access to off-site location		N/A
Ethical Approval from external organisations. If ethical approval from external organisations is pending give details below		N/A
Any other related documents		N/A

3. PROJECT DETAILS

a) Lay description (Maximum 200 words)

Please outline, in terms that any non-expert would understand, what your research project is about, including what participants will be required to do. Please explain any technical terms or discipline-specific phrases.

b) Research objectives (Maximum 150 words)

Please summarize briefly the objectives of the research,

To design, implement and evaluate a hybrid LSTM–XGBoost ensemble model that forecasts monthly Sri Lankan tourist arrivals more accurately than a classical SARIMA baseline, with particular emphasis on forecasting robustness during exogenous shock periods (the 2019 Easter attacks, the 2020–2021 COVID-19 pandemic, and the 2022 economic crisis). Specifically, the project will: (1) compile and clean a monthly dataset of tourist arrivals, exchange rates and search-trend indicators (2015–2025); (2) implement SARIMA, standalone LSTM and standalone XGBoost baseline models; (3) design a stacked hybrid LSTM–XGBoost ensemble; and (4) evaluate all models separately on stable and shock periods using RMSE, MAE, MAPE and the Diebold–Mariano test, supported by SHAP-based explainability.

Research location and duration

Locations / populations	Not applicable — no participants or physical research sites are involved. All data is sourced remotely from public online repositories (SLTDA, Central Bank of Sri Lanka, Google Trends).
Research start date	01/08/2026
Research end date	14/12/2026
Approximate duration	12 Weeks

4. PARTICIPANTS

Special Groups Consideration Table			
Do participants fall into any of the following special groups?	YES	NO	N/A
Minors (under 18 years of age)			N/A — no participants
People with learning or communication difficulties			N/A — no participants
Patients			N/A — no participants
People in custody			N/A — no participants
People engaged in illegal activities (e.g.)			N/A — no participants

5. SAMPLE DETAILS

Approximate number	Not applicable — no human sample; the dataset consists of monthly national statistics from 2015–2025 (approximately 120–132 monthly observations).
Where will participants be recruited from?	Not applicable — no participants are recruited. Data is retrieved directly from public sources (SLTDA, Central Bank of Sri Lanka, Google Trends).
Inclusion criteria	Not applicable
Exclusion criteria	Not applicable
Will participants be reunited and if so in what form?	Not applicable

Justification for proposed sample size and for selecting a specific gender, age, or any other group if this is done in your research.

Not applicable, as the research uses secondary aggregate statistical data rather than a human sample. The data time-frame (2015–2025) was chosen to include both stable periods and the three major exogenous shocks central to the research question.

6. RISKS TO PARTICIPANTS

a) Please describe any risks to participants that may arise due to the research. Such risks could include physical stress, emotional distress, perceived coercion e.g. lecturer interviewing own learners. Detail the measures and considerations you have put in place to minimize these risks

Not applicable — the project involves no human participants and therefore presents no physical, emotional, or psychological risk to any individual. The only risk-related consideration is at the level of responsible reporting: forecasts produced by the model will be presented with clearly stated confidence limitations, so that they are not misinterpreted as guaranteed predictions for policy or investment decisions by third parties who may later read the published report.

b) What will you communicate to participants about any identified risks? Will any information be withheld from them about the research purpose or procedure? If so, please justify this decision.

Not applicable

7. INFORMED CONSENT

Will you obtain active consent for participation?	N/A
Will you describe the main experimental procedures to participants in advance?	N/A
Will you inform the participants that their participation is voluntary and may be withdrawn at any point?	N/A
If the research is observational, will you ask for their consent to be observed?	N/A
With questionnaires, will you give participants the option of omitting questions that they do not want to answer?	N/A
Will you tell participants that their data will be treated with full confidentiality and that, if published, it will not be identifiable as theirs?	N/A
Will your project involve deliberately misleading participants in any way or will information be withheld?	N/A
If you answer yes, give details and justification for doing this.	N/A

8. DECLARATION

I have read and understand the NIBM guidelines for ethical practices in research and have read and understand the data protection guidelines.

Signed: _____ **Name:** _____

Date: _____

(Researcher)

Signed: _____ **Name:**

Date:

(Supervisor)

9. STATEMENT OF ETHICAL APPROVAL

Supervisor/Research Centre Director/Head of Department

This project has been considered using agreed procedures and is now approved/ referred to the Ethics Committee

Signed: _____ **Name:** _____

Date: _____

(Course Director)

Signed: _____ **Name:** _____

Date: _____

(Director of School)